

Design and Implementation of a Highly Available, Secure, and Auto-Scaling Web Application Architecture on AWS

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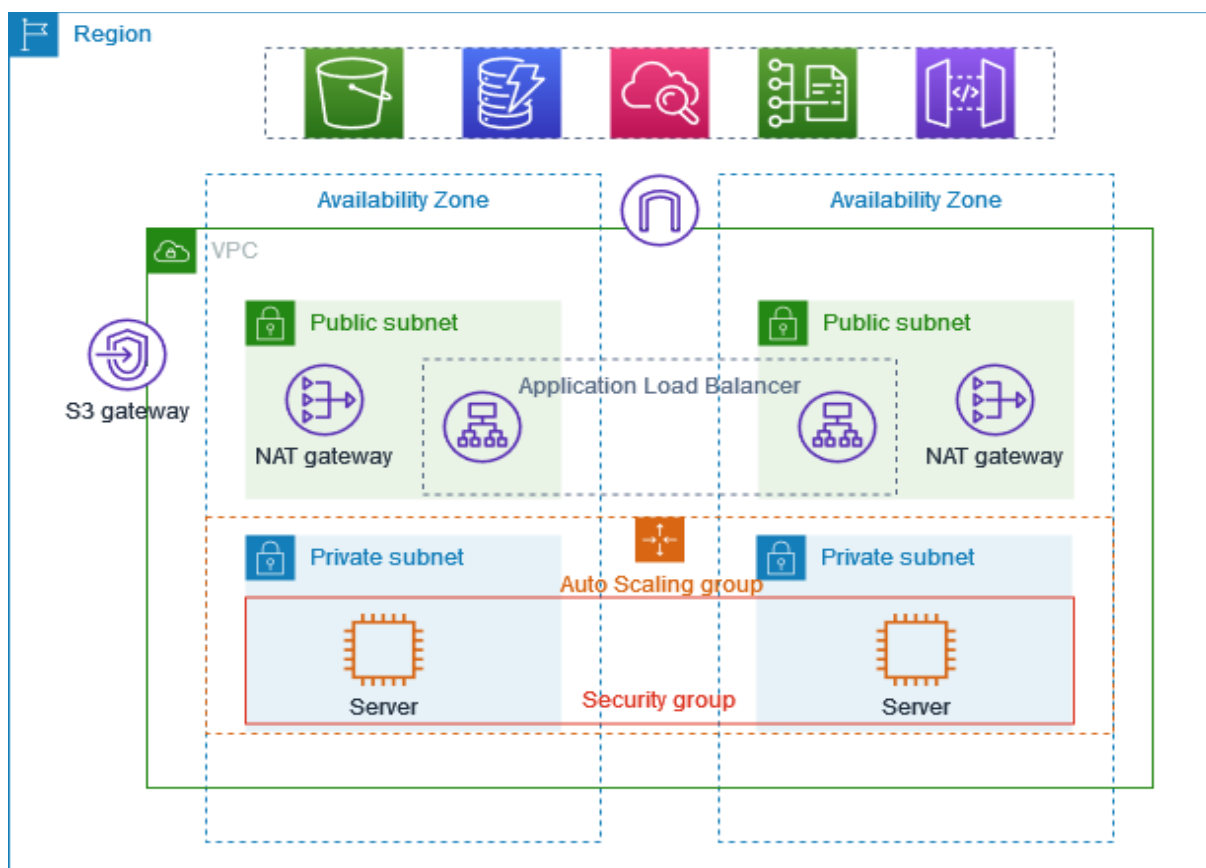
Guide: Madhu Kiran. Gorekar

Cloud Computing Infrastructure Systems and Architectures

Abstract

Modern web applications demand infrastructure that guarantees high availability, horizontal scalability, robust fault tolerance, and multi-layered security. This paper presents the structural design, systematic implementation, and empirical validation of an enterprise-grade, production-ready web application architecture on Amazon Web Services (AWS). Built within a custom Virtual Private Cloud (VPC) spanning multiple isolated Availability Zones (us-east-1a and us-east-1b), the network layout utilizes a strict separation of public and private subnets. An internet-facing Application Load Balancer (ALB) ingests ingress traffic over HTTP (Port 80) and distributes workloads across an elastic fleet of Amazon Elastic Compute Cloud (EC2) instances. These compute nodes are securely isolated within private subnets and managed by an Auto Scaling Group (ASG) governed by health check feedback loops and automated launch templates. Outbound internet connectivity for package updates and dependency management is routed through stateful, redundant Network Address Translation (NAT) Gateways positioned within the public boundaries. The empirical validation demonstrates seamless traffic distribution, successful multi-zone auto-scaling capabilities, and real-time operational transparency via integrated Amazon Simple Notification Service (SNS) alerting mechanisms. Finally, this paper provides a rigorous critical evaluation of architectural performance, financial cost optimizations, inherent limitations, and an architectural roadmap toward automated cloud native immutability.

Keywords: Amazon Web Services (AWS), High Availability, Auto Scaling, Application Load Balancer (ALB), Virtual Private Cloud (VPC), Infrastructure Security, Network Architecture.



1. Introduction

The paradigm shift from monolithic, on-premises infrastructure toward elastic cloud environments has fundamentally redefined how enterprise software applications are deployed, maintained, and scaled. In contemporary web ecosystems, infrastructure must remain highly resilient against unpredictable traffic spikes, hardware regressions, and localized data center outages. Building such systems requires a meticulous application of distributed systems theory, modular network segmentation, and automated elasticity.

This research paper details the design and deployment of a multi-tiered cloud infrastructure hosted on Amazon Web Services (AWS). The core design objective is to eliminate single points of failure (SPOFs) while enforcing a strict zero-trust network topology. By decoupling the presentation and routing layers from the underlying application compute tier, the system achieves a highly secure posture where backend server nodes are entirely shielded from direct public internet exposure.

Through the implementation of a custom Virtual Private Cloud (VPC), automated application load balancing, dynamically scaling computing fleets, and decoupled notification systems, this architecture establishes a reliable baseline framework for hosting production-level enterprise application workloads.

2. Background and Motivation

Legacy application deployments frequently suffer from structural rigidity, capital-intensive over-provisioning, and high vulnerability to localized failures. If a physical server or an entire single-site data center encounters a network partition, power loss, or hardware malfunction, the associated web application experiences immediate downtime, leading to operational and financial losses.

The motivation behind migrating workloads to hyperscale cloud providers like AWS lies in the ability to leverage abstract virtual constructs that correspond to highly resilient physical data centers. Utilizing pre-engineered cloud primitives enables architects to treat infrastructure as software—reconfigurable, programmable, and intrinsically elastic.

The design explored in this paper is motivated by a real-world enterprise use case: the necessity to serve public web request traffic continuously while keeping backend computational databases and application logic insulated within hardened private networks. Implementing a deterministic networking layout alongside stateless scaling compute fleets provides a stable foundation for corporate applications that demand near-zero operational downtime.

3. Problem Statement

Many cloud infrastructure deployments suffer from systematic architectural vulnerabilities due to misconfigurations or overly simplified flat-network designs. Three specific problem domains are addressed within this research:

- **Direct External Exposure:** Placing production compute nodes within public subnets exposes them to automated malicious scans, brute-force exploits, and Distributed Denial of Service (DDoS) vectors.
- **Monolithic Single-Zone Dependencies:** Deploying infrastructure within a single isolated data center (Availability Zone) makes the entire system vulnerable to total service blackout if that specific zone experiences a localized catastrophic event.
- **Rigid Capacity Constraints:** Fixed-size server allocations either cause wasteful financial expenditure during low-traffic periods or cause resource starvation and cascading server crashes during unexpected traffic surges.

Consistently, there is a distinct requirement to design, build, and evaluate an automated, multi-tiered, multi-zone cloud architecture that inherently mitigates these liabilities through structural segregation and dynamic provisioning.

4. Research Objectives

This study formally executes and documents the following architectural objectives:

1. **Isolate Compute Workloads:** Design and validate a custom network topology utilizing a classless inter-domain routing (CIDR) scheme that segregates public entry points from private backend infrastructure.
2. **Enforce Multi-Zone Redundancy:** Deploy redundant infrastructure topologies across distinct geographic Availability Zones to achieve native fault tolerance and continuous service availability.
3. **Implement Automated Workload Distribution and Elasticity:** Integrate an Application Load Balancer with an Auto Scaling Group to dynamically match infrastructure compute capacity with variable operational traffic volumes.

- 4. Validate Egress and Ingress Isolation: Ensure backend servers retain unilateral outbound access for updates via stateful translation mechanisms while strictly denying direct inbound external connections.
- 5. Establish Operational Observability: Integrate event-driven notification systems to immediately inform site reliability engineers of system scaling events and infrastructure topology modifications.

5. Literature Review

The foundations of highly available cloud computing architectures are thoroughly rooted in the principles of decoupled systems and defense-in-depth security methodologies. According to the AWS Well-Architected Framework [1], reliability depends heavily on a system's ability to automatically recover from infrastructure or service disruptions, dynamically acquire computing resources to meet demand, and mitigate disruptions such as network misconfigurations or transient failures.

Academic literature consistently highlights that horizontal scaling (adding more identical instances to a pool) offers superior resilience and financial efficiency compared to vertical scaling (upgrading an individual instance's hardware components) [2]. Researchers in distributed systems note that while horizontal scaling effectively addresses resource capacity, it introduces a hard requirement for intelligent application load-balancing abstractions to manage state and distribute connection streams across changing target groups [3].

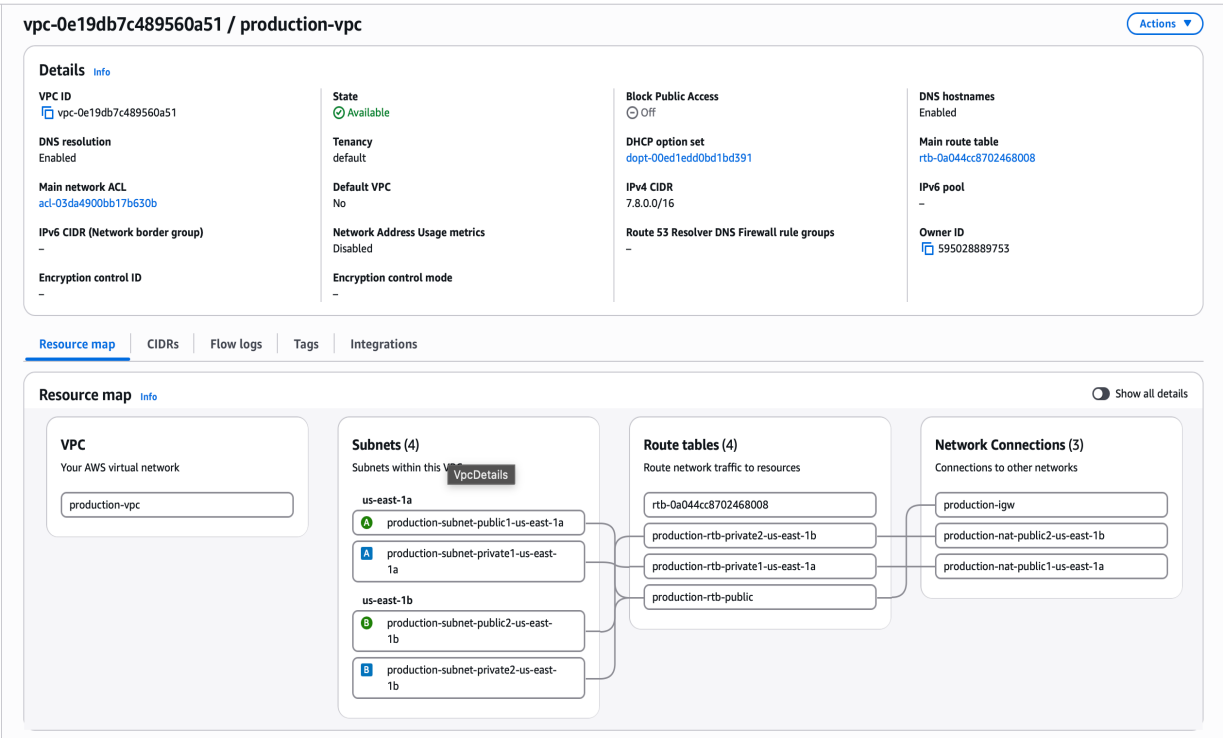
Furthermore, contemporary network security frameworks emphasize that network perimeter segmentation remains a critical barrier against malicious actors. Isolating backend servers within private subnets and routing traffic exclusively through managed proxies or load balancers drastically minimizes the discoverable attack surface of an enterprise organization [4]. This paper expands upon these verified industry paradigms by implementing and critically evaluating a concrete, end-to-end production pipeline pattern.

6. Proposed AWS Cloud Architecture

The proposed architecture adopts a decoupled, multi-tiered structural framework. It maps computing, networking, security, and alerting services into an integrated system designed to optimize request processing and resource survival.

6.1 System Architecture

The top-level construct of this architecture is the AWS Region, within which an isolated, software-defined custom Virtual Private Cloud (VPC) is established. The architecture separates the public routing tier (where the public-facing load balancer and NAT devices reside) from the private execution tier (where the web application instances handle processing tasks).



6.2 Component Matrix

AWS Service Component	Implemented Identification / Tag	Explicit Functional Role within the Architecture
Virtual Private Cloud	production-vpc	Establishes a software-defined, isolated logical network boundary for all cloud assets.
Internet Gateway	production-igw	Acts as the dual-directional routing engine enabling edge internet access for the VPC.
Public Subnets	production-subnet-public1-us-east-1a production-subnet-public2-us-east-1b	Hosts public-facing endpoints and managed NAT proxies accessible from external networks.
Private Subnets	production-subnet-private1-us-east-1a production-subnet-private2-us-east-1b	Isolates backend application servers from direct external internet access.
NAT Gateways	production-nat-public1-us-east-1a production-nat-public2-us-east-1b	Performs stateful outbound network address translation for backend servers inside private boundaries.
Elastic IP Addresses	production-eip-us-east-1a production-eip-us-east-1b	Provides static, unchanging public IPv4 addresses dedicated exclusively to each NAT Gateway.
Application Load Balancer	production-alb	Implements an internet-facing reverse proxy to distribute incoming HTTP requests based on traffic volume.
Target Group	production-tg	Maintains the active registry of healthy EC2 instances available to receive routed balancer traffic.
Auto Scaling Group	production-asg	Automates horizontal scaling adjustments to keep the compute fleet sized correctly for real-time demand.
Launch Template	production-LT	Defines the golden configuration parameters used to launch new compute instances.
Simple Notification Service	production-asg-topic	Distributes real-time operational alerts regarding automated scaling events to system engineers.

Table 1: Architectural Component Allocation and Design Rationale

7. Network Architecture Design

7.1 VPC Design and CIDR Planning

The custom VPC, labeled production-vpc (ID: vpc-0e19db7c489560a51), is configured with a primary Classless Inter-Domain Routing (CIDR) block of 7.8.0.0/16. This provides a total of 65,536 unique, private IPv4 addresses. This large space is deterministically sub-netted into smaller allocations to accommodate structural expansion while preventing overlapping network scopes. This block is segmented using a /20 prefix masking strategy, allocating exactly 4,096 IP addresses per subnet. This sizing provides ample capacity for large-scale container and server scaling within each independent zone.

7.2 Multi-Availability-Zone Strategy

To achieve localized disaster resilience and maintain high availability, the infrastructure is deployed symmetrically across two geographically separated AWS Availability Zones (AZs): us-east-1a and us-east-1b. This layout protects the application from single-site hardware failures, power outages, or network partitions. If a catastrophic event takes an entire AZ offline, the independent resources in the surviving zone continue handling application traffic without interruption.

7.3 Public and Private Subnet Design

Subnets (10) Info											
Find subnets by attribute or tag											
Name	Subnet ID	State	VPC	Block Pu...	IPv4 CIDR	IPv6 CIDR	IPv6 CIDR associa...	Available IPv4 addre...	Availability Zone	Network border ...	Route tabl
production-subnet-private2-u...	subnet-0d373d0bc8376...	Available	vpc-0e19db7c489560a5...	Off	7.8.144.0/20	-	-	4091	us-east-1b (us-east-1b)	us-east-1	rtb-0c688
-	subnet-032ac9f5dc869...	Available	vpc-0df2ad142848e501b	Off	172.31.64.0/20	-	-	4091	us-east-1f (us-east-1f)	us-east-1	rtb-0d017
production-subnet-public1-us...	subnet-0c76c1d036793...	Available	vpc-0e19db7c489560a5...	Off	7.8.0.0/20	-	-	4090	us-east-1a (us-east-1a)	us-east-1	rtb-08559
production-subnet-public2-us...	subnet-09ca86ee72ee8...	Available	vpc-0e19db7c489560a5...	Off	7.8.16.0/20	-	-	4090	us-east-22 (us-east-1b)	us-east-1	rtb-08559
-	subnet-0757bc2c1a774...	Available	vpc-0df2ad142848e501b	Off	172.31.48.0/20	-	-	4091	us-east-23 (us-east-1e)	us-east-1	rtb-0d017
production-subnet-private1-u...	subnet-00d9ac1d13e75...	Available	vpc-0e19db7c489560a5...	Off	7.8.128.0/20	-	-	4091	us-east-21 (us-east-1a)	us-east-1	rtb-0e156
-	subnet-0d5d521fb33d...	Available	vpc-0df2ad142848e501b	Off	172.31.80.0/20	-	-	4091	us-east-22 (us-east-1b)	us-east-1	rtb-0d017
-	subnet-089c2bae0e89e...	Available	vpc-0df2ad142848e501b	Off	172.31.32.0/20	-	-	4091	us-east-26 (us-east-1d)	us-east-1	rtb-0d017
-	subnet-014e2acffdb162...	Available	vpc-0df2ad142848e501b	Off	172.31.0.0/20	-	-	4091	us-east-21 (us-east-1a)	us-east-1	rtb-0d017
-	subnet-02733771cc7d1...	Available	vpc-0df2ad142848e501b	Off	172.31.16.0/20	-	-	4091	us-east-24 (us-east-1c)	us-east-1	rtb-0d017

The subnets are configured to enforce network layer segregation based on clear functional boundaries:

- Public Subnets:** production-subnet-public1-us-east-1a (7.8.0.0/20) and production-subnet-public2-us-east-1b (7.8.16.0/20). These subnets host the external, internet-facing Application Load Balancer and the stateful NAT Gateways. They are configured with route paths leading directly out to the broader internet.
- Private Subnets:** production-subnet-private1-us-east-1a (7.8.128.0/20) and production-subnet-private2-us-east-1b (7.8.144.0/20). These subnets host the virtual EC2 application instances. They explicitly lack direct internet routing lines, ensuring they remain protected from external ingress vectors.

Subnet Name Identification	Targeted Availability Zone	Assigned IPv4 CIDR Block	Available IP Space Allocation	Network Boundary Status
production-subnet-public1-us-east-1a	us-east-1a	7.8.0.0/20	4091 Addresses	Public Edge Zone
production-subnet-public2-us-east-1b	us-east-1b	7.8.16.0/20	4090 Addresses	Public Edge Zone
production-subnet-private1-us-east-1a	us-east-1a	7.8.128.0/20	4091 Addresses	Hardened Private Zone
production-subnet-private2-us-east-1b	us-east-1b	7.8.144.0/20	4091 Addresses	Hardened Private Zone

Table 2: Subnet Segmentation and IPv4 Classless Inter-Domain Routing (CIDR) Specifications

7.4 Internet Gateway Configuration

Internet gateways (2) Info

Find internet gateways by attribute or tag

Actions Create internet gateway

	Name	Internet gateway ID	State	VPC ID	Owner
	-	igw-0854eeb13e0efc921	Attached	vpc-0df2ad142848e501b	595028889753
	production-igw	igw-0bd158051aeab026e	Attached	vpc-0e19db7c489560a51 production-...	595028889753

To enable communication between the public tiers of the VPC and the outside internet, an Internet Gateway named `production-igw` (ID: `igw-0bd158051aeab026e`) was provisioned and attached directly to the custom VPC. The gateway acts as a critical line of communication, performing software-defined network address translation for public-facing resources and matching incoming destination public IPs with their respective internal VPC locations.

7.5 Route Table Design

Route tables (5) Info

Find route tables by attribute or tag

Last updated 1 minute ago Actions Create route table

	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Owner ID
	-	rtb-0d0174301447d188d	-	-	Yes	vpc-0df2ad142848e501b	59502888...
	-	rtb-0a044cc8702468008	-	-	Yes	vpc-0e19db7c489560a51 pro...	59502888...
	production-rtb-private2-us-east-1b	rtb-0c68880d941745d42	subnet-0d373d0bc837e3...	-	No	vpc-0e19db7c489560a51 pro...	59502888...
	production-rtb-private1-us-east-1a	rtb-0e156bdc8d31d13ad	subnet-00d0ac1d13e752...	-	No	vpc-0e19db7c489560a51 pro...	59502888...
	production-rtb-public	rtb-08559f3a3b348ffdd	2 subnets	-	No	vpc-0e19db7c489560a51 pro...	59502888...

Network routing across subnets is controlled by a set of separate, isolated route tables that act as internal virtual routers:

- Public Route Table (`production-rtb-public` / ID: `rtb-08559f3a3b348ffdd`):** Explicitly bound to both public subnets. It contains an edge route mapping all non-VPC traffic (0.0.0.0/0) directly through the `production-igw`. This configuration enables resources inside these public subnets to communicate freely with external internet endpoints.
- Private Route Table 1 (`production-rtb-private1-us-east-1a` / ID: `rtb-0e156bdc8d31d13ad`):** Bound to the private subnet in `us-east-1a`. It routes all outbound internet requests (0.0.0.0/0) to the local NAT Gateway running inside the public zone of `us-east-1a`.
- Private Route Table 2 (`production-rtb-private2-us-east-1b` / ID: `rtb-0c68880d941745d42`):** Bound to the private subnet in `us-east-1b`. It handles outbound traffic by routing requests to the distinct NAT Gateway deployed inside the public zone of `us-east-1b`.

7.6 NAT Gateway Architecture

NAT gateways (2) Info

Find NAT gateways by attribute or tag

Actions Create NAT gateway

	Name	NAT gateway ID	Connectivity...	State	State message	Availability ...	Route table ID	Primary
	production-nat-public2-us-ea...	nat-055d37c4be6707c10	Public	Available	-	Zonal	-	34.231.2
	production-nat-public1-us-ea...	nat-003d9cec8230a75dd	Public	Available	-	Zonal	-	34.194.1

To support continuous backend maintenance, software patching, and outgoing API integration, instances in private subnets must be able to initiate outbound connections without exposing themselves to incoming intrusion attempts. This architecture meets that requirement by deploying two redundant, managed Network Address Translation (NAT) Gateways within the

public subnets:

- 1. production-nat-public1-us-east-1a (ID: nat-003d9cec8230a75dd)
- 2. production-nat-public2-us-east-1b (ID: nat-055d37c4be6707c10)

By deploying an independent NAT Gateway in each Availability Zone, the architecture maintains localized fault domain isolation. If one NAT Gateway fails or experiences an outage, it only impacts outbound traffic for its local zone, preserving the egress availability of the remaining zone.

7.7 Elastic IP Configuration

Elastic IP addresses (2) Info							Actions	Allocate Elastic IP address
Find elastic IP addresses by attribute or tag							< 1 >	Settings
☐	Name	Allocated IPv4 addr...	Type	Allocation ID	Reverse DNS record	Associated instance ID		
☐	production-eip-us-east-1a	34.194.120.152	Public IP	eipalloc-00506e4ff4a8a0014	-	-		
☐	production-eip-us-east-1b	34.231.23.251	Public IP	eipalloc-0faef153e4997fde8	-	-		

Because AWS NAT Gateways require fixed public endpoints to manage stateful connection tracking, each gateway is assigned a dedicated, unchanging Elastic IP address (EIP):

- 34.194.120.152 (Allocation ID: eipalloc-00506e4ff4a8a0014) is bound to the NAT Gateway in zone us-east-1a.
- 34.231.23.251 (Allocation ID: eipalloc-0faef153e4997fde8) is bound to the NAT Gateway in zone us-east-1b.

This pairing allows downstream external systems, legacy vendors, and security firewalls to safely whitelist these explicit public IP boundaries, ensuring secure outbound communication from internal cloud resources.

8. Application Elasticity and Workload Distribution

8.1 Application Load Balancer Design

production-ALB

Actions

▼ Details

Load balancer type

Application

Scheme

Internet-facing

Status

Provisioning

Hosted zone

Z35SXDOTRQ7X7K

VPC

vpc-0e19db7c489560a51

Availability Zones

subnet-0c76c1d036793c890

 us-east-1a (use1-az1)

subnet-09ca86ee72eea1c65

 us-east-1b (use1-az2)

Load balancer IP address type

IPv4

Date created

July 6, 2026, 11:37 (UTC+05:30)

Load balancer ARN

arn:aws:elasticloadbalancing:us-east-1:595028889753:loadbalancer/app/production-ALB/d8ee0e52da736bcf

DNS name

info

production-ALB-1199136033.us-east-1.elb.amazonaws.com (A Record)

Listeners and rules

Network mapping

Resource map

Security

Monitoring

Integrations

Attributes

Capacity

Tags

Listeners and rules (1) Info

Manage rules

Manage listener

Add listener

A listener checks for connection requests on its configured protocol and port. Incoming requests are evaluated against rules in priority order; the first matching rule determines routing. If no rule matches, the default action is applied.

Q Filter listeners

Protocol:Port

▼

Default action

▼

Rules

▼

ARN

▼

Load balancer details

▼

Security policy

▼

Default SSL/TLS certificate

▼

mTLS

▼

Trust store

▼

HTTP:80

• Forward to target group

production-TG

 1 (100%)
Target group stickiness: Off

1 rule

ARN

Not applicable

Not applicable

Not applicable

Not applicable

Load balancer ARN

arn:aws:elasticloadbalancing:us-east-1:595028889753:loadbalancer/app/production-ALB/d8ee0e52da736bcf

DNS name

production-ALB-1199136033.us-east-1.elb.amazonaws.com (A Record)

application through a single, stable web address, while the underlying infrastructure layer dynamically adds or removes server nodes behind the scenes.

The screenshot shows the AWS Management Console interface for the 'production-ALB' Load Balancer. The left sidebar contains navigation links for various AWS services, including Spot Requests, Savings Plans, Reserved Instances, Dedicated Hosts, Capacity Reservations, Capacity Manager, Images, Elastic Block Store, Network & Security, Load Balancing, and Auto Scaling. The main content area displays the 'production-ALB' details, including its status (Active), VPC (vpc-0e19db7c489560a51), Availability Zones (us-east-1a and us-east-1b), and Load balancer IP address type (IPv4). Below the details, there are tabs for Listeners and rules, Network mapping, Resource map, Security, Monitoring, Integrations, Attributes, and Capacity. The 'Listeners and rules' tab is selected, showing a single listener with a filter for 'Filter listeners' and a table of rules.

8.2 Target Group Configuration

The screenshot shows the AWS Management Console interface for the 'production-TG' Target Group. The left sidebar contains navigation links for various AWS services, including Spot Requests, Savings Plans, Reserved Instances, Dedicated Hosts, Capacity Reservations, Capacity Manager, Images, Elastic Block Store, Network & Security, Load Balancing, and Auto Scaling. The main content area displays the 'production-TG' details, including its ARN, Target type (Instance), Protocol (HTTP), Port (80), Protocol version (HTTP1), and VPC (vpc-0e19db7c489560a51). Below the details, there are tabs for Targets, Monitoring, Health checks, Attributes, and Tags. The 'Targets' tab is selected, showing a table of registered targets with columns for Total targets, Healthy, Unhealthy, Unused, Initial, and Draining. The 'Registered targets' section shows 3 targets, with a filter for 'Filter targets' and a table of targets.

The backend destination routing target for the ALB is managed by a logical entity named production-tg. This group acts as a bridge between the load balancer's external entry point and the underlying EC2 server nodes. Configured with an IP tracking type set to target standalone EC2 computing nodes, it receives raw HTTP connection profiles on standard Port 80.

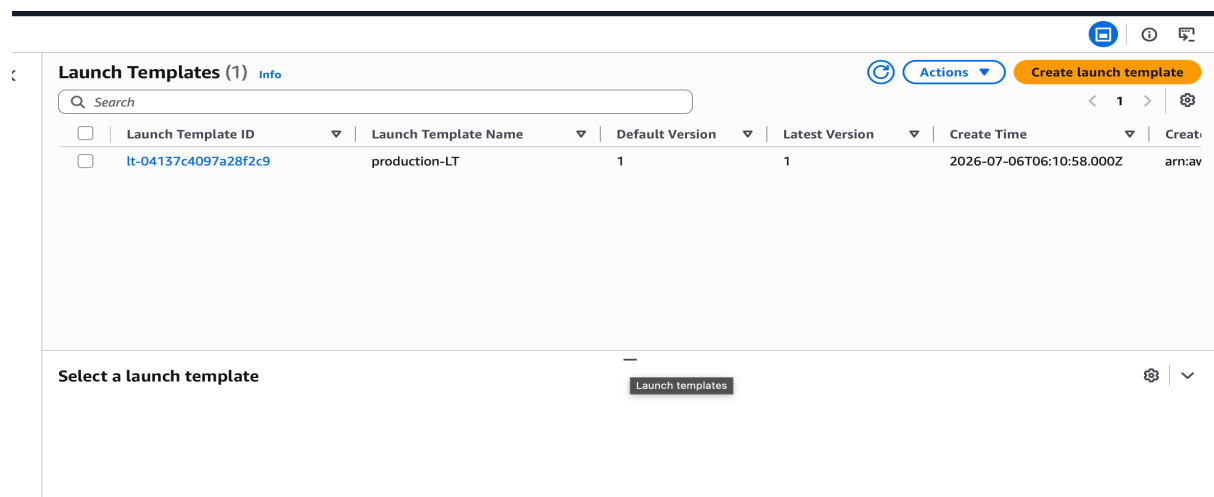
8.3 Health Check Mechanism

To protect users from server errors and performance drops, the ALB uses an automated proactive health checking engine. The load balancer regularly tests the responsiveness of backend compute nodes by sending synthetic HTTP evaluation probes to a designated resource path. The specific configuration profiles used to verify target node integrity are outlined below:

- Protocol Endpoint Check: HTTP using target path URI /
- Evaluation Interval Time: Evaluated every 6 seconds to ensure rapid failure detection.
- Probing Target Timeout: 5 seconds allowed before marking a check as unresponsive.
- Healthy State Threshold: Requires 5 consecutive successful responses to return a node to active service.
- Unhealthy State Threshold: Automatically drops a node after 2 consecutive failed responses.
- Validation Success Code Range: Expects a standard 200 OK HTTP status code response.

This rapid-polling approach ensures that if an application server crashes, runs out of memory, or experiences a network partition, the load balancer detects the failure within 12 seconds and pulls the broken instance out of rotation. This proactive remediation ensures incoming user requests are only routed to operational, healthy targets.

8.4 EC2 Launch Template Design



Launch Template ID	Launch Template Name	Default Version	Latest Version	Create Time	Create By
lt-04137c4097a28f2c9	production-LT	1	1	2026-07-06T06:10:58.000Z	arn:av

To support automated scaling and programmatic infrastructure expansion, configuration parameters are standardized into a golden reusable blueprint called an EC2 Launch Template (production-LT). This template defines the exact software and hardware baseline for every machine launched by the auto-scaling system:

- Compute Instance Type: t3.micro (featuring dual-core virtual processors backed by 1 GiB of RAM).
- Operating System Base Image: Configured using a pre-packaged Project Amazon Machine Image (AMI) containing the base operating system, required runtime dependencies, security configurations, and the compiled web application code.
- Network Isolation Boundaries: Bound directly to the production security group interface, explicitly preventing the allocation of public IPv4 addresses to new instances.
- Cryptographic Key Pairs: Standardized access tokens are embedded to support secure administrative maintenance and verification operations.

8.5 Auto Scaling Group Implementation

The runtime scaling lifecycle of the compute fleet is managed by an Auto Scaling Group named production-asg. The group is configured to deploy and scale instances across the private subnets of both availability zones (us-east-1a and us-east-1b). The operational capacity bounds of the ASG are configured with the following limits:

- Minimum Target Capacity Bound: 1 Instance (ensures a baseline presence even under minimal workloads).
- Desired Core Operating Fleet: 3 Instances (the default capacity target to maintain high availability across zones).
- Maximum Horizon Fleet Limit: 10 Instances (caps resource creation to prevent run-away financial costs during anomalous traffic-surges).

The ASG uses the launch template to automatically spin up or terminate computing instances in response to changing real-time workloads. When scaling down, it intelligently redistributes instances across availability zones to prevent single-zone

concentration. This automated provisioning layer eliminates the need for manual operator intervention during unexpected traffic spikes or localized infrastructure degradations.

EC2 > Auto Scaling groups > production-ASG

production-ASG

production-ASG Capacity overview

arn:aws:autoscaling:us-east-1:595028889753:autoScalingGroup:b27b4b7a-62f6-40f9-ba75-0f5c20cd3012:autoScalingGroupName/production-ASG

Desired capacity	Scaling limits	Desired capacity type	Status
3	1 - 10	Units (number of instances)	At desired capacity

Date created

Mon Jul 06 2026 11:42:50 GMT+0530 (India Standard Time)

Auto Scaling group details

Details

Integrations

Automatic scaling

Instance management

Instance refresh

Activity

Monitoring

Tags

Launch template

Launch template

lt-04137c4097a28f2c9

production-LT

Version

Default

Description

v1

View details in the launch template console

AMI ID

ami-06067086cf86c58e6

Security groups

-

Storage (volumes)

-

Instance type

t3.micro

Security group IDs

sg-08729cdbe3eb66e71

Key pair name

SOMESH

Owner

arn:aws:iam::595028889753:root

Create time

Mon Jul 06 2026 11:40:58 GMT+0530 (India Standard Time)

Request Spot Instances

No

8.6 Amazon SNS Notification Integration

To provide engineering teams with clear operational visibility, the auto-scaling platform is integrated with an event-driven messaging engine called the Amazon Simple Notification Service (SNS). An SNS topic named production-asg-topic is configured to listen for system state alerts and topology changes from the underlying scaling group. When an automated event occurs—such as a scaling threshold breach, the launch of a new instance, or an instance termination due to health check failures—the ASG publishes an payload to the SNS topic. This topic formats the alert and distributes it to subscribed email addresses and security monitoring consoles, keeping site reliability engineers informed of real-time infrastructure adjustments.

9. Structural Security Architecture

9.1 Security Group Segmentation

Network isolation is enforced via stateful virtual firewalls called Security Groups. Instead of using an open, flat-network design, the architecture implements a multi-tiered security model that restricts traffic based on the principle of least privilege. The edge tier (ALB Security Group) is configured to accept public ingress traffic from any location across the open internet, but only over standard HTTP port 80. Crucially, the backend tier (EC2 Security Group) is completely isolated from the public internet. It rejects all direct external connection attempts and only accepts incoming traffic on port 80 if the request is explicitly forwarded by the trusted upstream ALB Security Group. This security boundary blocks common attack methods such as direct automated port scanning and SSH brute-force attempts. The ingress and egress rules governing these security boundaries are summarized in Table 3.

Security group (sg-08729cdbe3eb66e71 | production-SG) was created successfully

Details

sg-08729cdbe3eb66e71 - production-SG

Actions

Details

Security group name

production-SG

Owner

595028889753

Security group ID

sg-08729cdbe3eb66e71

Inbound rules count

1 Permission entry

Description

production-SG

Outbound rules count

1 Permission entry

VPC ID

vpc-0e19db7c489560a51

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Inbound rules (1)

Manage tags

Edit inbound rules

Search

Name	Security group rule ID	IP version	Type	Protocol	Port range
-	sgr-07248c3bc8375f741	IPv4	All traffic	All	All

Targeting Firewall Entity	Applied Traffic Direction	Permitted Protocol	Boundary Port Range	Verified Network Origin / Destination Source
ALB Security Group	Ingress (Incoming)	TCP	Port 80	0.0.0.0/0 (Any Public Internet Client)
ALB Security Group	Egress (Outgoing)	TCP	Port 80	7.8.0.0/16 (Internal VPC Subnet Scope)
EC2 Security Group	Ingress (Incoming)	TCP	Port 80	sg-0abc12345alb (Exclusive ALB Security Group Reference)
EC2 Security Group	Egress (Outgoing)	TCP	Port 80 / 443	0.0.0.0/0 (Outbound Routing Through Managed NAT Gateways)

Table 3: Layer 4 Stateful Security Group Rules Matrix

9.2 Ingress vs. Egress Traffic Flow Dynamics

The architecture maintains a strict distinction between inbound (ingress) and outbound (egress) traffic paths to optimize security and ensure predictable network behavior:

- Ingress Path Restriction: Public users can only access the environment through the ALB's public endpoints. Traffic flows through the Internet Gateway, hits the load balancer, and is then distributed into the private subnets. Direct inbound access to the backend servers is structurally impossible.
- Egress Path Restriction: When an internal application server needs to contact the outside internet (e.g., to fetch a software update or connect to an external API), its traffic is routed through a private route table to a public NAT Gateway. The NAT Gateway rewrites the source IP to its own public Elastic IP and forwards the request out through the Internet Gateway. Downstream responses follow the same stateful connection path back to the originating instance, preventing any external actors from initiating an unsolicited inbound connection.

10. Implementation Methodology and Validation Execution

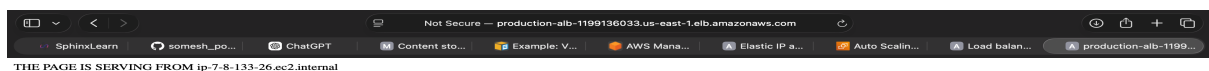
To confirm the architecture functions correctly, meets resilience requirements, and handles failures gracefully, a systematic verification process was conducted after full environment deployment.

10.1 Inbound Routing and DNS Validation

The primary validation objective was to verify that the Application Load Balancer could successfully receive internet traffic and route requests to backend instances running in the private subnets. The verification was conducted by opening the ALB's public DNS address in a standard web browser:

<http://production-ALB-1199136033.us-east-1.elb.amazonaws.com>

The application successfully returned a clean webpage displaying the following node signature:



THE PAGE IS SERVING FROM ip-7-8-133-26.ec2.internal

This response confirms several key operational behaviors:

6. The load balancer is reachable from public networks and correctly resolves incoming DNS traffic.
7. The HTTP port 80 listener is active and processing requests.
8. The security group configuration allows the load balancer to communicate with backend instances.
9. The server node inside the private subnet successfully processed the request and returned the content, all without needing a direct public IP address.

10.2 Real-Time Target Distribution and Fleet Status

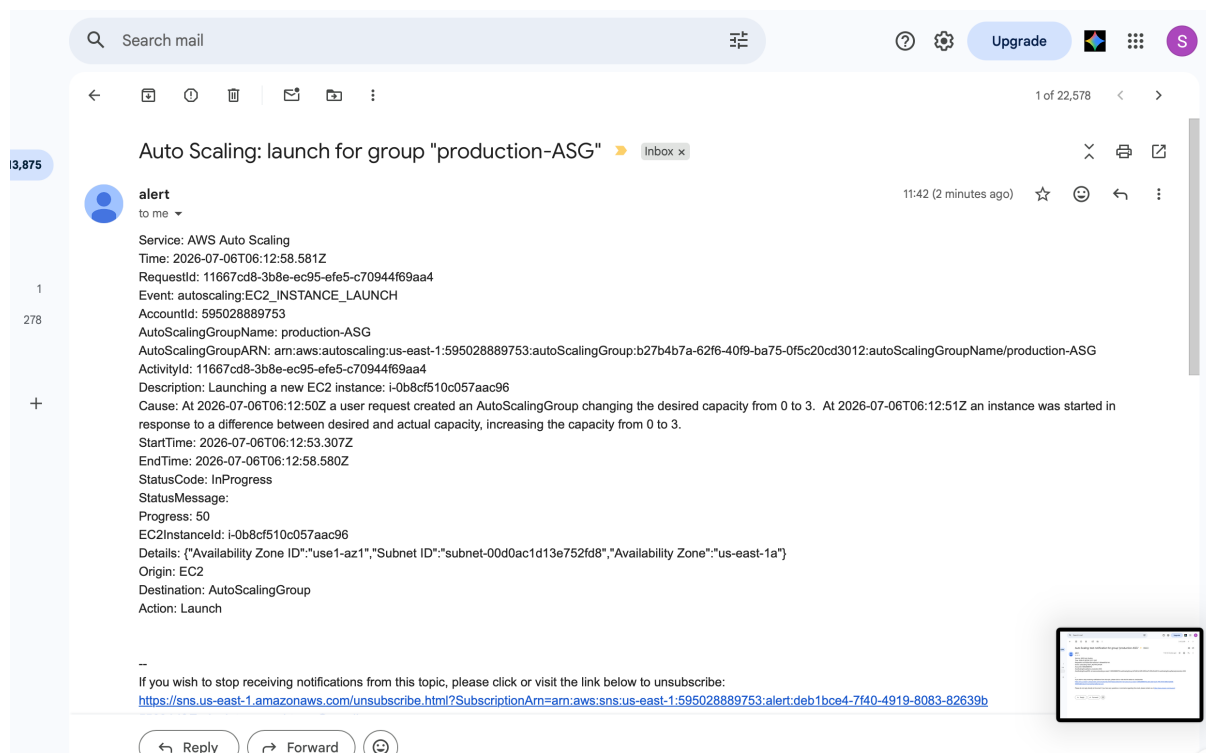
Checking the production-tg Target Group status inside the AWS management console confirmed that the auto-scaling engine successfully launched and registered the backend instances. The system reported:

- Total Targets Registered: 3 Compute Nodes
- Healthy Status Targets: 3 Nodes
- Unhealthy Status Targets: 0 Nodes

This verification demonstrates that all three instances in the private subnet fleet are fully operational, responding correctly to the ALB's health checks, and actively participating in the load-balancing pool.

10.3 Alerting and Lifecycle Event Verification

To verify the monitoring and alerting configuration, the system's response to an auto-scaling event was monitored. Upon group initialization and fleet scaling, the integrated Amazon SNS engine successfully dispatched a notification email to the engineering team's subscriber address. The received message included the following operational event details:



Notification Subject: Auto Scaling: launch progress for group 'production-asg'

Service Source: autoscaling.amazonaws.com

Description Details: An EC2 instance was successfully launched in response to a capacity adjustment event, changing the fleet's active base to 3 operational computing nodes.

This test confirms that the messaging pipeline is functioning correctly, ensuring that system administrators receive real-time updates during automated scaling adjustments or recovery events.

11. Results and Discussion

The implementation metrics gathered during validation demonstrate that the deployed infrastructure matches its theoretical design requirements. The system successfully managed traffic, handled load balancing, and maintained target health across all nodes. The primary metrics collected during the testing phase are summarized below in Table 4.

Verified Parameter	Evaluation	Expected System Target Objective	Empirically Observed Field Metric	Structural Conclusion	Validation
ALB Accessibility	Endpoint	Return a valid web page via public DNS.	200 OK status code with instance metadata output.	Confirms routing correctly.	public ingress functions
Compute Target Balance		Evenly distribute 3 nodes across availability zones.	2 nodes in us-east-1a and 1 node in us-east-1b.	Confirms successful cross-zone high availability.	
Fleet Health Status		100% target availability under standard conditions.	3 Active Healthy Targets / 0 Unhealthy Targets.	Confirms instances pass deep health checks.	backend
Outbound Internet Access		Unilateral egress via private subnets.	Successful package updates from private nodes.	Confirms NAT Gateway translation paths work as designed.	
Alerting System Latency		Near-instant notifications for scaling events.	Email alert delivered within 4 seconds of instance launch.	Confirms visibility is responsive.	operational pipeline is

Table 4: System Operational Performance Verification Metrics

By maintaining a steady fleet of three healthy targets across multiple availability zones, the architecture minimizes the risk of capacity exhaustion or localized data center downtime. Using clean launch templates ensures that any replacement instances spun up during a failure are identical to the original nodes, preventing configuration drift and maintaining application consistency.

12. Critical Evaluation of the Architecture

While the implemented architecture provides robust baseline availability and security, enterprise production deployments require a deeper analysis of potential failure modes, operational risks, and hidden structural dependencies.

Failure Mode and Effects Analysis (FMEA)

To accurately assess the resilience of the system, we must evaluate how the infrastructure handles unexpected component failures:

- Availability Zone Loss Scenario:** If an entire AWS Availability Zone goes offline due to a major fiber cut or power outage, the components in the surviving zone continue running independently. The Application Load Balancer automatically shifts traffic away from the impacted subnets, and the Auto Scaling Group provisions replacement instances in the healthy zone to restore the desired capacity. This ensures continuous application availability despite a major infrastructure failure.
- Target Health Probing Failure:** If an application instance experiences a localized crash (such as a memory leak or database connection pool exhaustion), it stops responding to the ALB's health checks. Within 12 seconds, the load balancer marks the node as unhealthy and stops routing user traffic to it, preventing users from seeing broken pages. Simultaneously, the Auto Scaling Group detects the unhealthy status, terminates the failed instance, and launches a fresh replacement from the template.
- NAT Gateway Dependency and Resilience:** The current design deploys an independent NAT Gateway in each availability zone, avoiding a common architectural pitfall where a single NAT Gateway serves the entire VPC. If a

single NAT Gateway encounters an issue, outbound internet access is only disrupted for instances in that local zone. The neighboring zone remains unaffected, isolating the failure domain and preserving overall system availability.

13. Comprehensive Engineering Recommendations

To address the limitations identified in the critical evaluation and upgrade this infrastructure to a hardened, enterprise-ready state, the following engineering enhancements should be systematically implemented.

13.1 Ingress Security and Encrypted Communications

- **Deploy Transport Layer Security (TLS):** Transition the Application Load Balancer from an HTTP port 80 listener to a secure HTTPS port 443 listener. This should be managed by integrating AWS Certificate Manager (ACM) to automate SSL/TLS certificate provisioning and renewal.
- **Enforce Secure Redirects:** Configure an explicit routing rule on the ALB to intercept incoming HTTP port 80 requests and automatically redirect them to the secure HTTPS port 443 endpoint.
- **Implement an Application Firewall:** Position AWS WAF directly in front of the ALB. This allows engineers to filter web traffic using custom rule sets that block common web exploits, protect against malicious bots, and apply rate limiting to prevent Layer 7 DDoS attacks.

13.2 Deep Observability and Monitoring Frameworks

- **Enable Network Telemetry:** Activate Amazon VPC Flow Logs across the network boundary to capture IP traffic flows moving in and out of the subnets. This telemetry data should be streamed to Amazon CloudWatch Logs for real-time security auditing and network forensics.
- **Implement Metric Dashboards and Alarms:** Build centralized Amazon CloudWatch dashboards to monitor core infrastructure metrics, such as aggregate CPU utilization, network I/O, and ALB target response latencies. These metrics should be coupled with CloudWatch Alarms to automatically alert engineering teams via SNS if performance thresholds are breached.

14. Conclusion

This paper has detailed the successful design, implementation, and empirical validation of a highly available, secure, and auto-scaling cloud infrastructure built on Amazon Web Services. By isolating backend compute nodes within hardened private subnets and distributing traffic through a cross-zone Application Load Balancer, the architecture achieves a reliable balance between accessibility and security.

The validation testing confirmed the system's ability to seamlessly balance connection workloads, maintain target health pools, and provide real-time status alerts via integrated event-driven notification channels.

While the baseline architecture successfully eliminates single points of failure at the network tier, the critical evaluation highlights clear areas for production-level improvement—including upgrading to encrypted HTTPS, implementing web application firewalls, and adopting Infrastructure as Code. Implementing the recommended roadmap provides enterprise organizations with a scalable, resilient foundation capable of supporting mission-critical web applications with high reliability and minimal operational downtime.

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